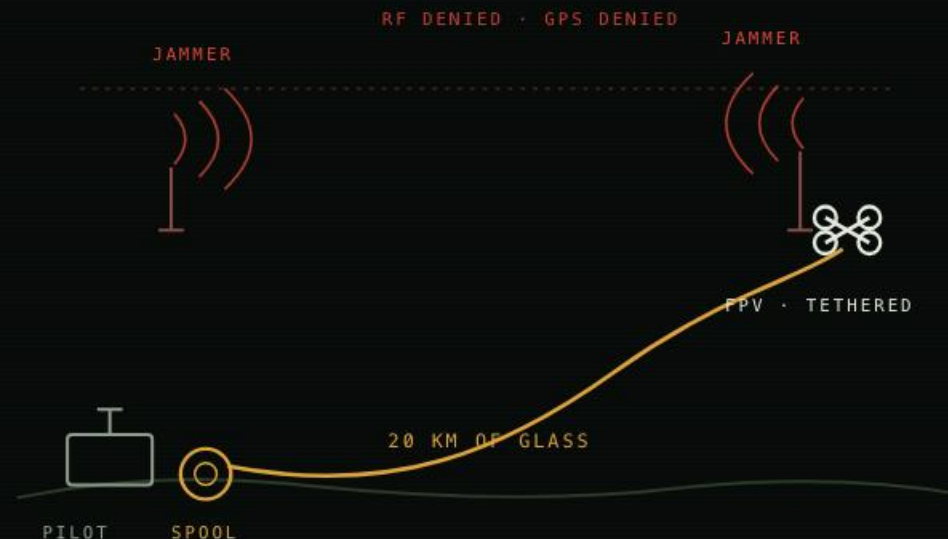


— VESPER · DN HACKS

The drone war is a comms war.

And right now, the state of the art is **a leash.**

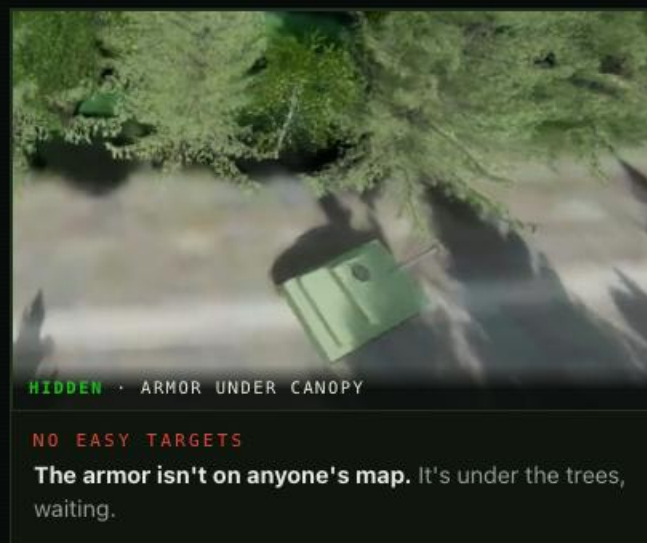
EASHAN VAGISH · THEO OLTEAN



UKRAINE, TODAY: CREWS FLY STRIKE DRONES ON 20 KILOMETERS OF UNSPOOLING GLASS — BECAUSE THE SPECTRUM IS TOO JAMMED FOR RADIO. ORIGINAL ILLUSTRATION.

— THE PROBLEM

Out there, anything that phones home **dies.**



And the line that never moves: **a human pulls the trigger. Every time.**

KRAMATORSK_MS · 2000 M x 2000 M, N UP

live t=23 s · found 1/3 · reached 0/3 · search.pt

500 m

— TRACK ♦ WAYPOINTS ▲ SIGHTED ● REACHED ◊ UNFOUND

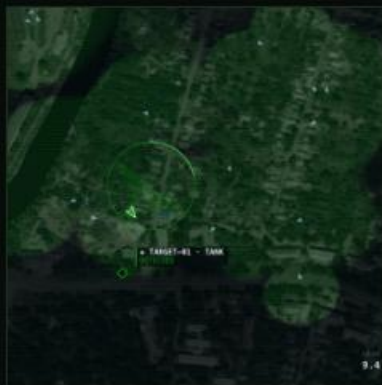
drag to pan · scroll to zoom · click a marker

You can't scale a cord.
We cut it — with software.

Sixteen drones that hunt in radio silence — and everything they need fit a laptop we could carry into the field. No datacenter. No datalink. No leash.

— THE HUNT

Find it. Call it in. Get the nod. Kill it.



01

SEARCH

One shared map. Sixteen hunters sweeping it.



02

DETECT

Armor in the treeline. Tagged in seconds.



03

UPLINK

Jammed? Hold the report. Fly to where the link lives.



04

APPROVE

The request hits your screen. You approve — or you don't.



05

STRIKE

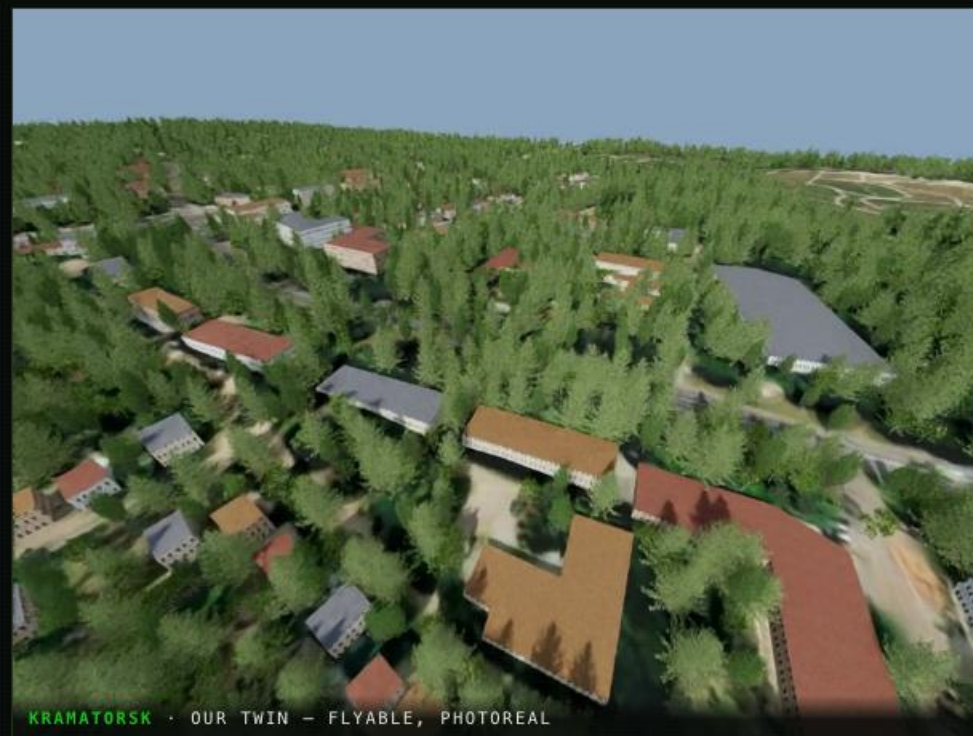
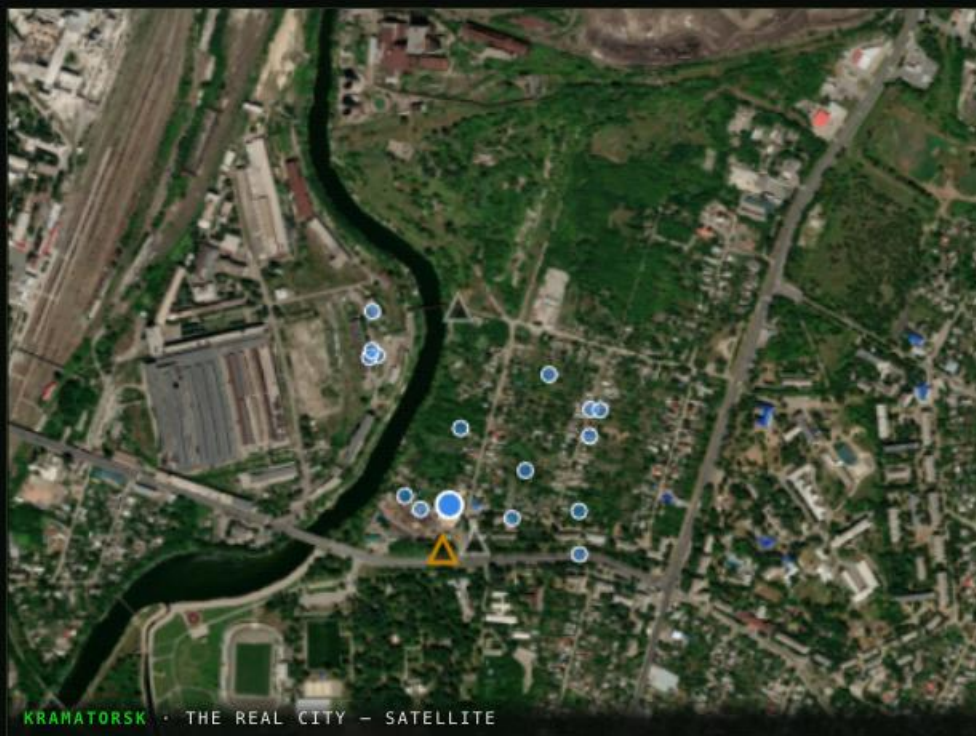
One nod. One munition. One kill.

It hunts alone. It never kills alone.

— HOW · THE TWIN

We cloned the battlefield.

Give us coordinates. We pull public satellite and terrain data and rebuild the city in 3D — **every road, roof, and tree** — a digital twin the swarm can die in, ten thousand times a night.



— HOW · THE REPS

A million sorties **before the first real one.**



LOST ———> METHODICAL ———> **LETHAL** · SAME POLICY, SAME TERRAIN — THE ONLY THING THAT CHANGED IS TRAINING

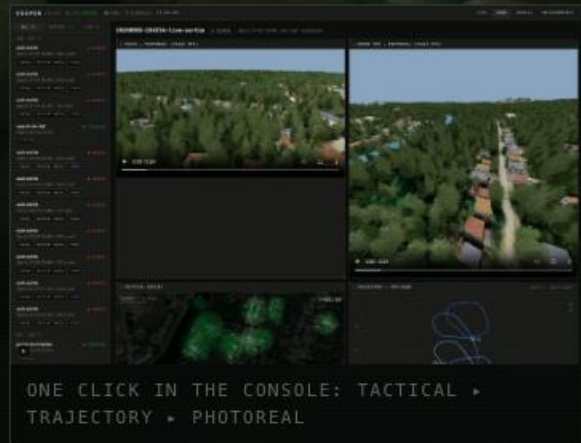
Millions of rollouts in the twin. Nobody wrote a waypoint. Nobody scripted a behavior. **Reward did the teaching — the kill lives in the weights.**

— AFTER ACTION

Then fly it back — through the drone's eyes.

Every sortie replays photoreal in Isaac Sim — the same flight, frame for frame, from search to strike.

ISAAC RTX REPLAY OF THE RECORDED STRIKE SORTIE — SIMULATION RENDER, NOT COMBAT FOOTAGE.



— VESPER

Point it at any coordinates **on Earth.**

Kramatorsk today. Anywhere tomorrow. The twin builds itself — the swarm trains overnight.

EDGE-AUTONOMOUS — RUNS ON A MAC **COMMS**-AWARE, JAM-RESILIENT **HUMAN**-AUTHORIZED, ALWAYS **RL**-TRAINED, NOT SCRIPTED

Eashan Vagish
ev264@cornell.edu

Theo Oltean
tio3@cornell.edu

The cord is cut.